

Question Number	Answer	Mark
7(a)(i)	The only correct answer is D a solid ball of totipotent cells <i>A is not correct because a morula is not a hollow balls of cells</i> <i>B is not correct because a morula is not a hollow balls of cells</i> <i>C is not correct because a morula contains totipotent cells</i>	(1)

Question Number	Answer	Additional guidance	Mark
7(a)(ii)	An answer that makes reference to the following points: (source of) {fatty acids / glycerol / lipids / amino acids / substrate / glucose / energy / ATP} for {respiration / metabolic reactions / protein synthesis / growth of embryo / development of embryo / new cells / mitosis}	candidates may refer to morula / blastocyst / zygote / embryo ignore nutrients / food accept a source or store of something relevant and then a relevant reason why Please note energy & ATP could be in LHS or RHS bracket e.g. energy for mitosis / lipids as a source of energy	(1)

Question Number	Answer	Additional guidance	Mark
7(b)	A description that makes reference to five of the following points: • differential gene expression occurred (1) • only genes needed for skeletal muscle cell formation are {active / switched on} (1) • due to {epigenetic modification / histone modification / DNA methylation} (1) • only {active / switched on/ skeletal muscle} genes are transcribed (into mRNA) (1) • translation (of mRNA) occurs (at the ribosome) (1) • (resulting in) formation of proteins needed in {skeletal muscle cells / mitochondria / muscle contraction} (1)	accept muscle cells for skeletal muscle cells accept genes not required in skeletal muscle cells are switched off accept transcription factors accept description e.g. acetyl group added to histone accept correct references to splicing of pre-mRNA accept responses which merge mp3 and mp4 together correctly e.g. acetylation of the gene allows it to be transcribed accept named {proteins / structures} in skeletal muscle cells accept proteins produced (permanently) modify the cell into a skeletal muscle cell	(5)

Question Number	Answer	Additional guidance	Mark
7(c)(i)	A calculation in which: • reading value from graph and subtracting (1) • calculation of number of cells in interphase	<u>Example of calculation:</u> $100 - 5 = 95$ or $5\% \text{ of } 2000 = 100$ $(95 \div 100) \times 2000 = 1900$ 1832 = 1 mark (for correct calculation done with OVCAR5 value) ecf for wrong number from graph between 4.8 to 5.2 = 1896 to 1904 Correct answer scores full marks	(2)

These are the most common credit worthy responses – but please follow the MS at all times for other responses

1	2
100	1900
95	
1832	
1896	
1897	
1898	
1899	
1901	
1902	
1903	
1904	

Question Number	Answer	Mark
7(c)(ii)	The only correct answer is A HOC8 <i>B is not correct because OVCAR5 does not have the highest mitotic index</i> <i>C is not correct because OVCAR8 does not have the highest mitotic index</i> <i>D is not correct because SKOV3 does not have the highest mitotic index</i>	(1)

Question Number	Answer	Additional guidance	Mark
7(c)(iii)	An explanation that makes reference to the following points: (use of embryonic stem cells) means destruction of an embryo (1) (which some people consider to be) an {ethical / moral} issue (1) IMP	accept (use of embryonic stem cells) means the {embryo / potential life} is killed ignore harmed ignore embryo is a potential life unqualified accept the candidate's view that it is unethical / morally wrong accept the embryos cannot give consent accept {for religious reasons / against their beliefs}	(2)